AEROTRACK IoT — TOTAL SYSTEM ARCHITECTURE

End-to-End Edge-to-Cloud RFID Baggage Tracking & Automated Cargo Hold Reconciliation | UPF IoT Project

ESP32-S2 + ESP-IDF v6.1 + MQTT

1. PHYSICAL RAMP ENVIRONMENT

Aircraft Cargo Door & Conveyor Belt

Smart Luggage Tag

(Real Adhesive Sticker)

- 13.56 MHz HF (ISO 14443A)
- NTAG213 / Mifare Classic 1K
- Unique 7-Byte UID / Serial
- Affixed to Passenger Suitcase

13.56 MHz RF Link

Physical Ramp Inputs

(Simulated via Potentiometers)

- Potentiometer 1 (P1): Conveyor Belt Loading Speed
- Potentiometer 2 (P2): Cargo Hold Hatch Angle (Open vs. Closed Door)

Analog 0-3.3V

Visual Ground Feedback

Ground Handler Work Area

- Green: Valid Bag Ingested
- Red: Misrouted Alert / Halt
- Blue: Cloud Network Status

Visual Signaling

2. EDGE HARDWARE & FIRMWARE NODE

Espressif ESP32-S2 (Xtensa 32-bit LX7 @ 240 MHz)

Hardware Peripherals & Pinout

RC522 RFID Module

- 4-Wire SPI Master
- SCK=18, MISO=19
- MOSI=23, SS=5, RST=22
- 13.56 MHz RF Field

ADC & Status LEDs

- 13-bit SAR ADC (P1, P2)
- GPIO1 (Belt), GPIO2 (Door)
- RGB LEDs / RMT Driver
- Non-blocking signaling

ESP-IDF v6.1 Firmware Architecture (FreeRTOS)

Task: rfid_scanner_task (Priority 5)

Continuous SPI interrogation, anti-collision, 2s debounce filter

Task: sensor_telemetry_task (Priority 3)

ADC oneshot sampling (P1, P2), moving average filter

Task: cloud_mqtt_task (Priority 4)

esp_mqtt_client (QoS 1), auto-reconnect, JSON serialize

Task: actuator_led_task (Priority 2) - Non-blocking LED sequences

Pub

Sub

3. TRANSPORT & NETWORK

Wireless Telemetry & Commands

Wi-Fi 802.11 b/g/n

- Airport 2.4 GHz Access Point
- DHCP / Static IP
- WPA2/WPA3 Enterprise

MQTT over TLS

Port 8883 / Port 1883

- OASIS MQTT Standard
- Guaranteed Delivery: QoS 1
- Keep-Alive & Auto-Reconnect
- Lightweight JSON Payloads

Tópicos de Mensajería

Uplink (ESP32 → Cloud):

- /luggage/scan (UID, Speed)
- /telemetry (ADC, RSSI)

Downlink (Cloud → ESP32):

- /commands (HOLD_LOCK)
- /manifest/sync

TLS Link

4. CLOUD SERVICES & USER INTERFACES

Azure IoT Hub F1 / AWS IoT Core / Backend Services

Cloud IoT Ingestion Gateway

Azure IoT Hub (F1 Free Tier) / AWS IoT Core

- TLS Termination & Token/Certificate Authentication
- Device Twin & Status Synchronization
- Ingestion Throughput & Telemetry Routing Engine

Cargo Hold Reconciliation Engine

Automated Tally & Validation Service

- Flight Manifest Database (e.g. 200 registered check-in bags)
- Real-Time Hold Tally Counter (199 loaded, 1 missing)
- Mismatch Detection: Flags bags meant for another flight
- IATA Resolution 753 Audit Logger & Compliance Trail

Airline Ground Dashboard

(Ramp Operator Portal)

- Live Hold Tally Gauge
- Conveyor Speed Dial
- Missing Bag Alert Banner
- Door Close Confirmation

Passenger Tracking App

(Web & Mobile Client)

- 'Bag In Hold' Notification
- Multi-Leg Transit Status (BCN → BER → SYD)
- Reassurance on Transfer